values for fluoride are national averages and should be used with caution when levels of fluoride in locally produced foods and beverages are of interest or concern.

***Vitamins*.** Vitamins included in the database are ascorbic acid (vitamin C), thiamin, riboflavin, niacin, pantothenic acid, vitamin B_6, vitamin B_{12}, added vitamin B_{12}, folate, total choline, betaine, vitamin A (individual carotenoids, and retinol), vitamin E (tocopherols and tocotrienols), added vitamin E, vitamin K (phylloquinone, dihydrophylloquinone and menaquinone-4), and vitamin D (D_2 and D_3).

Ascorbic acid. In the current database system, nearly all data for ascorbic acid listed under Nutrient No. 401, total ascorbic acid, have been determined by the microfluorometric method (AOAC 967.22). Older values which have not been updated are primarily for reduced ascorbic acid and were determined by the dichloroindophenol method (AOAC 967.21).

Thiamin, Riboflavin, and Niacin. Thiamin is determined chemically by the fluorometric method (AOAC 942.23). Fluorometric (AOAC 970.65) or microbiological (AOAC 940.33) methods are used to measure riboflavin. Niacin is determined by microbiological methods (AOAC 944.13). The values for niacin are for preformed niacin only and do not include the niacin contributed by tryptophan, a niacin precursor. The term “niacin equivalent” applies to the potential niacin value; that is, to the sum of the preformed niacin and the amount that could be derived from tryptophan (the mean value of 60 mg tryptophan is considered equivalent to 1 mg niacin (IOM, 1998)). Although not included in SR, niacin equivalents can be estimated for those foods where amino acids are given:

mg Niacin equivalents = mg niacin + (mg tryptophan / 60).

Pantothenic acid, Vitamins B_6, and B_{12}. Pantothenic acid (AOAC 945.74 or 992.07), vitamin B_6 (AOAC 961.15), and vitamin B_{12} (AOAC 952.20) are determined by microbiological methods. Vitamin B_{12} is found intrinsically in foods of animal origin or those containing some ingredient of animal origin, e.g., cake that contains eggs or milk. For foods that contain only plant products, the value for vitamin B_{12} is assumed to be zero with the exception of some fortified foods discussed below. Some reports contain values for vitamin B_{12} in certain fermented foods (soy sauce and miso). It is believed that this B_{12} is synthesized not by the microorganisms responsible for the fermentation of the food, but rather by other contaminating microorganisms. Therefore, one should not consider these foods to be a consistent source of vitamin B_{12} (Liem *et al*., 1977) and these values are not included in the database.

The Dietary Reference Intakes (DRI) report on vitamin B_{12} recommended that people older than 50 years meet their Recommended Dietary Allowances (RDA) mainly by consuming foods fortified with vitamin B_{12} or a vitamin B_{12}-containing supplement (IOM, 1998). Since vitamin B_{12} added as a fortificant may provide a significant source of the vitamin in the diet, Nutrient No. 578 for “added vitamin B_{12}” has been added to the database. In this release, there are over 300 foods fortified with vitamin B_{12}. The vast

majority are breakfast cereals, infant formulas, and plant-based meat substitutes. For these foods, the value for total vitamin B_{12} is used for "added vitamin B_{12}." Only a few cereals containing a milk ingredient would contain any intrinsic vitamin B_{12}. Milk-based infant formulas should contain intrinsic vitamin B_{12}. Plant-based meat substitutes should not contain intrinsic vitamin B_{12}.

Folate. Values are reported for folic acid (Nutr. No. 431), food folate (Nutr. No. 432), and total folate reported in µg (Nutr. No. 417) and as dietary folate equivalents (DFEs) (Nutr. No. 435). These varied folate forms are included and defined as described in the DRI report on folate (IOM, 1998). RDAs for folate are expressed in DFEs, which take into account the greater bioavailability of synthetic folic acid compared with naturally occurring food folate.

To calculate DFEs for any single food, it is necessary to have separate values for naturally occurring food folate and added synthetic folic acid in that item.

µg DFE = µg food folate + (1.7 * µg folic acid)

Folate values for foods analyzed through NFNAP are generated using the trienzyme microbiological procedure (Martin *et al*., 1990). For a small number of foods, total folate was determined as the sum of one or more individual folate vitamers (5-methyltetrahydrofolate, 10-formyl folic acid, 5-formyltetrahydrofolic acid, and tetrahydrofolic acid); these are indicated in the footnotes. Microbiological methods measure µg total folate; for enriched foods, folic acid and food folate are not distinguished from each other. Therefore, to be able to calculate DFE, multi-ingredient enriched foods are analyzed by an additional microbiological procedure without enzymes to estimate the amount of added folic acid (Chun *et al*., 2006). Food folate is then calculated by difference.

The addition of folic acid to enriched cereal-grain products subject to standards of identity began in the United States on January 1, 1998 (CFR, Title 21, Pts. 136—137). These products include enriched flour, cornmeal and grits, farina, rice, macaroni, noodles, bread, rolls, and buns. Folic acid may continue to be added (with some restrictions on amounts) to breakfast cereals, infant formulas, medical foods, food for special dietary use, and meal replacement products.

For unenriched foods, food folate would be equivalent to total folate since folic acid (pteroylmonoglutamic acid) occurs rarely in foods. Therefore, the same value with its number of data points and standard error, if present, is used for total folate and food folate. The folic acid value is assumed to be zero.

For enriched cereal-grain products with standards of identity (flour, cornmeal and grits, farina, rice, macaroni, noodles, bread, rolls, and buns), the folic acid value is calculated by subtracting the analytical folate value before fortification from the analytical value for the fortified product.

Some ready-to-eat (RTE) cereals have been fortified with folic acid for over 25 years, and food folate values (before fortification) are not readily available for these products. Food folate was estimated by means of the NDBS formulation program for a variety of high-consumption cereals. Mean folate values were calculated for categories of RTE cereals based on grain content. Added folic acid was then calculated by subtracting estimated food folate from the total folate content. Generally, food folate values represent a small proportion of the total folate in the fortified products.

Choline and Betaine. Beginning with SR19 (2006), total choline and betaine values from the USDA Database for the Choline Content of Common Foods, Release 2 (USDA, 2008) have been incorporated into SR. In some cases, newer values have been incorporated into SR and these supersede the values in the Special Interest Database for choline. Values for the individual metabolites have not been added to SR, but are available in the USDA Database for the Choline Content of Common Foods, (http://www.ars.usda.gov/Services/docs.htm?docid=6232).

For analysis, choline compounds are extracted, partitioned into organic and aqueous phases using methanol and chloroform, and analyzed directly by liquid chromatography-electrospray ionization-isotope dilution mass spectrometry (LC-ESI-IDMS) (Koc *et al*., 2002). Samples are analyzed for betaine and these choline-contributing compounds: free choline (Cho), glycerophosphocholine (GPC), phosphocholine (Pcho), phosphatidylcholine (Ptdcho), and sphingomyelin (SM).

Because there are metabolic pathways for the interconversion of Cho, GPC, Pcho, PtdCho, and SM (Zeisel *et al*., 1994), total choline content is calculated as the sum of these choline-contributing metabolites. Betaine values are not included in the calculation of total choline since the conversion of choline to betaine is irreversible (Zeisel *et al*., 2003).

Vitamin A. Beginning with SR15 (2002), values for vitamin A in µg of retinol activity equivalents (RAEs) and µg of retinol are reported. At the same time, values in µg of retinol equivalents (REs) were dropped from the database.

This change responded to new reference values for vitamin A in the DRI report issued by the Institute of Medicine of the National Academies (IOM, 2001). The report recommended changing the factors used for calculating vitamin A activity from the individual provitamin A carotenoids and introduced RAE as a new unit for expressing vitamin A activity. One µg RAE is equivalent to 1 µg of all-*trans*-retinol, 12 µg of all-*trans*-β-carotene, or 24 µg of other provitamin A carotenoids. The RAE conversion factors are based on studies showing that the conversion of provitamin A carotenoids to retinol was only half as great as previously thought.

Vitamin A is also reported in international units (IU), and will continue to be reported because it is still the unit used for nutrition labeling in the US. One IU is equivalent to 0.3 µg retinol, 0.6 µg β-carotene, or 1.2 µg other provitamin-A carotenoids (NAS/NRC, 1989) and thus over-estimates bioavailability.

Individual carotenoids (β-carotene, α-carotene, β-cryptoxanthin, lycopene, and lutein+zeaxanthin) are reported. The analytical data are from NFNAP, generated using HPLC methodology (AOAC 941.15 or Craft, 2001). Most analytical systems do not separate lutein and zeaxanthin, so these carotenoids are shown combined. These values supersede those in Holden *et al*., 1999. Vitamin A activity values in RAE and IU were calculated from the content of retinol and individual carotenoids (β-carotene, α-carotene, and β-cryptoxanthin) using the appropriate factors. For food items used in the FNDDS, carotenoid values are imputed if analytical data are not available. For many of these items data are only available for vitamin A in IU. The variability in carotenoid levels due to cultivar, season, growing area, etc., as well as rounding within the NDBS, increases the difficulty in matching the calculated vitamin A activity values from imputed individual carotenoids to the existing IU values. As a result, the vitamin A IU value agrees within ±15 IU of the value calculated from individual carotenoids.

When individual carotenoids are not reported for plant foods (i.e., fruits, vegetables, legumes, nuts, cereal grains, and spices and herbs), µg RAE are calculated by dividing the IU value by 20. In foods of animal origin, such as eggs, beef, pork, poultry, lamb, veal, game, and fish (except for some organ meats and dairy), all of the vitamin A activity is contributed by retinol. For these foods, where analytical data are not available, µg RAE and µg of retinol are calculated by dividing the IU value by 3.33.

In foods that contain both retinol and provitamin A carotenoids, the amount of each of these components must be known to calculate RAE. Previously, most of the vitamin A data in the database were received as IU. Therefore, the amounts of the provitamin A carotenoids and retinol were then estimated from the ingredients. Once the components had been estimated, µg RAE were calculated as (IU from carotenoids/20) + (IU from retinol/3.33). Micrograms of retinol were calculated as IU from retinol/3.33.

Vitamin D. Due to considerable public health interest in vitamin D, a multi-year project was undertaken by NDL to expand and update the relatively small existing dataset of vitamin D values in SR. Much of the original data for vitamin D had been published earlier in USDA's Provisional Table (PT-108) (Weihrauch and Tamaki, 1991), with values for fortified foods updated as needed with data received from the food industry. Earlier data collected between 1999 and 2008 utilized AOAC methods 982.29 or 992.26.

The availability of vitamin D data for foods permitting subsequent dietary intake assessment is expected be a useful tool in investigating dietary requirements of vitamin D in vulnerable groups, one of the specific research recommendations of the 2005 Dietary Guidelines Committee (DGAC. 2004). An Institute of Medicine Committee to Review Dietary Reference Intakes for Vitamin D and Calcium was convened in 2009 to assess current relevant data and revise, as appropriate, the DRIs for vitamin D and calcium. Their report was issued in 2011 (IOM).

Cholecalciferol (vitamin D_3; Nutr. No. 326) is the form naturally occurring in animal products and the form most commonly added to fortified foods. Ergocalciferol (vitamin D_2; Nutr. No. 325) is the form found in plants and is added to some fortified foods, such as soy milk. In SR27, vitamin D (Nutr. No. 328) is defined as the sum of vitamin D_2 and vitamin D_3.

Before foods could be analyzed for vitamin D and included in SR, analytical methodology had to be developed that could be used for a variety of food matrices (Byrdwell, 2008). Although a single method is not required for USDA-sponsored analyses, all participating laboratories must demonstrate that their analysis of quality control materials falls within an acceptable range of values. For vitamin D, all methods involved extraction with solvent(s), cleanup steps, and quantification by HPLC or by HPLC and LC/MS. In the absence of certified quality control materials for vitamin D, NDL, in collaboration with Virginia Tech, developed five matrix-specific materials, one of which was sent with every batch of foods to be analyzed. The materials were: vitamin D_3 fortified fluid milk, a vitamin D_3 fortified multigrain ready-to-eat cereal, orange juice fortified with calcium and vitamin D_3, pasteurized process cheese fortified with vitamin D_3, and canned red salmon, a natural source of D_3 (Phillips *et al*., 2008). Vitamin D may also be present as 25-hydroxycholecalciferol in some foods such as fish, meat, and poultry. At this point the analytical methodology used to determine this metabolite of vitamin D has not been sufficiently validated; when work on this validation is completed 25-hydroxycholecalciferol values will be provided in future releases of SR.

Once an improved method of analysis was developed (Byrdwell, 2008), and the laboratories certified, a selection of foods, representing natural vitamin D sources and fortified sources, were chosen for sampling and analysis under the NFNAP (Haytowitz *et al*., 2008). Analyses have been completed for raw eggs and the following fortified products: fluid milk at 4 fat levels, reduced fat chocolate milk, fruit yogurt, and orange juice. Current analytical values for fish are based on limited analyses; additional samples are being analyzed and values will be updated in future SR releases. Vitamin D analyses have also been completed for selected cuts/pieces of chicken, pork, and beef. These data have been determined by a new LC/MS/MS method (Huang and Winters, 2011).

Vitamin D values in SR27 are provided in both micrograms (µg) and International Units (IU) to support both the analytical unit (µg) and the unit (IU) that is currently used in nutrient labeling of foods in the US. The biological activity of vitamin D is given as 40 IU/µg. Where available, specific isomers of vitamin D are reported only in µg. Calculations for vitamin D in SR include:

Vitamin D, µg (Nutr. No. 328) = vitamin D_2, µg + vitamin D_3, µg
Vitamin D, IU (Nutr. No. 324) = vitamin D, µg x 40

Vitamin D values in µg (Nutr. No. 328) are provided for all items in SR27 used to create the FNDDS.

In some cases, it was possible to identify food groups for which the foods do not provide or only contain trace amounts of vitamin D. Values for those foods were set to zero. For example, except for mushrooms, plant foods are not expected to contain any appreciable levels of vitamin D. In order to provide vitamin D estimates for the rest of the foods provided to create the FNDDS, data for other foods have been taken from the scientific literature or from other food composition databases, calculated from industry-declared % DV fortification levels, determined by formulation/recipe techniques, or estimated by other USDA imputation methods.

Fluid milk available at the retail level is fortified. The dairy industry provided guidance that most dairy products used as ingredients in formulated (commercial multi-ingredient) foods, are not likely to be fortified with vitamin D. Likewise, margarine used in commercial products is generally not vitamin D-fortified; a relatively low number of vitamin D-fortified margarines and spreads are available in the retail market. For ingredients that could be fortified at the retail level, but generally are not fortified at the food processing level, two related profiles are available in SR – one with added vitamin D and one without. When estimates were calculated for formulated foods, the unfortified profile was used. For home-prepared foods, such as pudding prepared with milk, the fortified ingredient(s) was selected for use in the recipe calculation of vitamin D. In the case of margarine, a market-share blend of fortified and unfortified product was used.

For some retail products, such as yogurt, there is considerable brand-to-brand difference in vitamin D fortification practices. One brand or line of products may be fortified with vitamin D, whereas another brand may not. Both types are included in the database. The market changes quickly and consumers concerned about vitamin D intake should always confirm vitamin D content by checking the product label.

Vitamin E. The DRI report (IOM, 2000) defines vitamin E as the naturally occurring form (*RRR*-α-tocopherol) and three synthetic forms of α-tocopherol. Since the release of SR16-1 (2003), NDL has reported vitamin E as mg of α-tocopherol (Nutr. No. 323) in accordance with the DRI report. Analytical values for tocopherols found in the database are determined by gas-liquid chromatography (GLC) or high-performance liquid chromatography (HPLC; Ye *et al*., 2000). Although β, γ, and δ-tocopherol do not contribute to vitamin E activity, they are included in the database when analytical data are available and starting with SR27 (2013) data on α-, β, γ, and δ-tocotrienol have also been included.

In the 2000 DRI report, a revised factor was recommended for calculation of the milligram amounts of α-tocopherol contributed by synthetic forms of vitamin E, since *all rac*-α-tocopherol contains 2*R*-stereoisomeric and 2*S*-stereoisomeric forms in equal amounts. Vitamin E activity to establish recommended intakes is limited to the 2*R*-stereoisomeric forms of α-tocopherol (IOM, 2000).

However, the unit for vitamin E required by the NLEA is IU and is based on the 1968 RDA definitions for vitamin E (CFR, Title 21, Pt. 101) (US Food and Drug Administration–Department of Health and Human Services, 2004).

When NDL receives vitamin E data from the food industry expressed as IU, the values are converted to mg amounts based on the conversions of vitamin E in IU to mg as defined by the DRI report:

One mg of α-tocopherol = IU of the *all rac*-α-tocopherol compound × 0.45; and
One mg of α-tocopherol = IU of the *RRR*-α-tocopherol compound × 0.67.

The basis of the vitamin E tolerable upper intake level (UL), another important reference value defined in the DRI report, was established using all forms of supplemental α-tocopherol (IOM, 2000). Although the 2*S*-stereoisomers do not contribute to dietary requirements for vitamin E (IOM, 2000), they do contribute to the total intake relative to the UL. Starting with SR18 (2005), "added vitamin E" (Nutr. No. 573) was added to the database. In this release, there are about 140 food items that have values for added vitamin E greater than 0. For the majority of these food items, the form added is *all rac*-α-tocopherol; these values should be multiplied by 2 to relate intakes of this form to the UL. Items that are fortified with *RRR*-α-tocopherol are identified by a footnote and the added vitamin E value can be used directly to estimate its contribution to the UL.

Vitamin K. Much of the data for vitamin K has been generated under NFNAP and supersedes the values in the USDA Provisional Table (PT-104) (Weihrauch and Chatra, 1994). Vitamin K is extracted with hexane, purified with solid phase extraction using silica columns, and quantitated using HPLC with chemical reduction and fluorescence detection. Losses are corrected using vitamin $K_{1(25)}$ as the internal standard (Booth *et al*., 1994). Starting with SR23 (2010), in addition to data on vitamin K_1 (Nutr. No. 430), data on dihydrophylloquinone (Nutr. No. 429) and menaquinone-4 (Nutr. No. 428) are also released. Dihydrophylloquinone is created during the commercial hydrogenation of plant oils. Menaquinone-4 is formed from vitamin K_1 and/or the synthetic form of vitamin K found in animal feed, and is found primarily in meats and meat products.

***Lipid Components*.** Fatty acids are expressed as the actual quantity of fatty acid in g per 100 g of food and do not represent fatty acids as triglycerides. Historically, most fatty acid data were obtained as the percentage of fatty acid methyl esters and determined by GLC analyses (AOAC 996.06). These data were converted to g fatty acid per 100 g total lipid using lipid conversion factors and then to g fatty acid per 100 g edible portion of food using the total lipid content. Details of the derivation of lipid conversion factors were published by Weihrauch *et al*., (1977).

In the redesigned NDBS, fatty acid data may be imported in a variety of units and converted within the system. No conversions are required if data are received as g fatty acid per 100 g edible portion of food. Data received as fatty acid esters and as triglycerides are converted to fatty acids using Sheppard conversion factors. Sheppard

conversion factors are based on the molecular weights of the specific fatty acid and its corresponding esters (butyl or methyl) and triglyceride (Sheppard, 1992). When fatty acid data are received as percentages of fatty acid methyl esters, methyl esters are converted to fatty acids using Sheppard conversion factors and then multiplied by total lipid (Nutrient No. 204) to give g fatty acid per 100 g edible portion of food. Occasionally, total lipid values are available from a variety of data sources, but individual fatty acids are available from fewer sources. In those cases, it may be necessary to normalize the individual fatty acids to the mean fat value of the food item. In the case of normalized fatty acids, the sum of the individual fatty acids will equal the mean fat value multiplied by the Weihrauch (1977) lipid conversion factor for that food item. No statistics of variability are reported for normalized fatty acids.

Individual Fatty Acids. The basic format for describing individual fatty acids is that the number before the colon indicates the number of carbon atoms in the fatty acid chain and the number after the colon indicates the number of double bonds. For unsaturated fatty acids, additional nutrient numbers have been added to accommodate the reporting of many specific positional and geometric isomers. Of the specific isomers, there are two basic classifications considered: omega double bond position and *cis/trans* configuration of double bonds.

Omega-3 (n-3) and omega-6 (n-6) isomers are denoted in shorthand nomenclature as n-3 and n-6. The n- number indicates the position of the first double bond from the methyl end of the carbon chain. The letter *c* or *t* indicates whether the bond is *cis* or *trans*. For polyunsaturated fatty acids, *cis* and *trans* configurations at successive double bonds may be indicated. For example, linoleic acid is an 18 carbon omega-6 fatty acid with 2 double bonds, both in *cis* configuration. When data are isomer specific, linoleic acid is described as 18:2 n-6 *c,c.* Other isomers of 18:2, for which nutrient numbers have now been assigned, include 18:2 *c,t*; 18:2 *t,c*; 18:2 *t,t;* 18:2 *t* not further defined; and 18:2 *i.* 18:2 *i* is not a single isomer but includes isomers other than 18:2 n-6 *c,c* with peaks that cannot be easily differentiated in the particular food item. Systematic and common names for fatty acids are given in Table 2.

Table 2 is provided for the convenience of users in attaching common names or systematic names to fatty acids in this database. Though individual fatty acids are more specific than in past releases, it is not possible to include every possible geometric and positional isomer. Where specific isomers exist for a fatty acid, the common name of the most typical isomer is listed for the undifferentiated fatty acid and an asterisk (*) designates the specific isomer to which that name applies. For example, the most typical isomer for 18:1 is oleic acid. Thus, undifferentiated 18:1 is designated in Table 2 as oleic acid, but an asterisk by 18:1 c indicates the common name for 18:1 oleic acid actually only applies to this isomer.

Table 2.—Systematic and Common Names for Fatty Acids

Fatty acid	Systematic name	Common name of most typical isomer	Nutrient number
Saturated fatty acids			
4:0	butanoic	butyric	607
6:0	hexanoic	caproic	608
8:0	octanoic	caprylic	609
10:0	decanoic	capric	610
12:0	dodecanoic	lauric	611
13:0	tridecanoic		696
14:0	tetradecanoic	myristic	612
15:0	pentadecanoic		652
16:0	hexadecanoic	palmitic	613
17:0	heptadecanoic	margaric	653
18:0	octadecanoic	stearic	614
20:0	eicosanoic	arachidic	615
22:0	docosanoic	behenic	624
24:0	tetracosanoic	lignoceric	654
Monounsaturated fatty acids			
14:1	tetradecenoic	myristoleic	625
15:1	pentadecenoic		697
16:1 undifferentiated	hexadecenoic	palmitoleic	626
16:1 *cis*			673*
16:1 *trans*			662
17:1	heptadecenoic		687
18:1 undifferentiated	octadecenoic	oleic	617
18:1 *cis*			674*
18:1 *trans*			663
20:1	eicosenoic	gadoleic	628
22:1 undifferentiated	docosenoic	erucic	630
22:1 *cis*			676*
22:1 *trans*			664
24:1 *cis*	cis-tetracosenoic	nervonic	671
Polyunsaturated fatty acids			
18:2 undifferentiated	octadecadienoic	linoleic	618
18:2 *trans* not further defined			665
18:2 *i* (mixed isomers)			666
18:2 n-6 *cis, cis*			675*
18:2 *trans, trans*			669
18:2 conjugated linoleic acid (CLAs)			670
18:3 undifferentiated	octadecatrienoic	linolenic	619
18:3 n-3 *cis, cis, cis*		alpha-linolenic	851*
18:3 n-6 *cis, cis, cis*		gamma-linolenic	685

Fatty acid	Systematic name	Common name of most typical isomer	Nutrient number
18:3 *trans* (other isomers)			856
18:3 *i* (mixed isomers)			866
18:4	octadecatetraenoic	parinaric	627
20:2 n-6 *cis, cis*	eicosadienoic		672
20:3 undifferentiated	eicosatrienoic		689
20:3 n-3			852
20:3 n-6			853
20:4 undifferentiated			620
20:4 n-6	eicosatetraenoic	arachidonic	855*
20:5 n-3	eicosapentaenoic (EPA)	timnodonic	629
21:5			857
22:4			858
22:5 n-3	docosapentaenoic (DPA)	clupanodonic	631
22:6 n-3	docosahexaenoic (DHA)		621

* Designates the specific isomer associated with the common name; the typical isomer is listed for the undifferentiated fatty acid.

Fatty acid totals. Only a small portion of the fatty acid data received for release in SR27 contains specific positional and geometric isomers. Therefore, it has been necessary to maintain the usual nutrient numbers corresponding to fatty acids with no further differentiation other than carbon length and number of double bonds. To aid users of our data, specific isomers are always summed to provide a total value for the undifferentiated fatty acid. For example, mean values for the specific isomers of 18:2 are summed to provide a mean for 18:2 undifferentiated (Nutrient No. 618). Other fatty acid totals provided are: (1) the sum of saturated, monounsaturated, and polyunsaturated fatty acids; and (2) the sum of *trans*-monoenoic, the sum of *trans*-polyenoic, and the sum of all *trans* fatty acids.

Values for total saturated, monounsaturated, and polyunsaturated fatty acids may include individual fatty acids not reported; therefore, the sum of their values may exceed the sum of the individual fatty acids. In rare cases, the sum of the individual fatty acids may exceed the sum of the values given for the total saturated fatty acids (SFA), monounsaturated fatty acids (MUFA), and polyunsaturated fatty acids (PUFA). These differences are generally caused by rounding and should be relatively small.

For multi-ingredient processed brand-name foods, industry data are often available for fatty acid classes (SFA, MUFA, and PUFA) but are lacking for individual fatty acids. In these cases, individual fatty acids are calculated from the fatty acids of the individually listed ingredients and normalized to the total fat level. A best-fit approximation has been made to fatty acid classes but, unavoidably, calculated sums of individual fatty acid totals do not always match industry data for fatty acid classes. Zero values for individual fatty acids should be understood to mean that trace amounts may be present. When g